Full Implementation of BaSyx Asset Administration Shell Infrastructure: From Web UI to Real-Time Data Integration

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Abstract

This comprehensive study details the deployment and configuration of an industrial Digital Twin platform utilizing the Eclipse BaSyx framework. The research covers the end-to-end implementation including the React-based Web UI, a Dockerized AAS environment, and a persistent storage mechanism using MongoDB Atlas. We investigate the integration of a BaSyx DataBridge to facilitate real-time telemetry from IoT devices using MQTT and JSONata transformations. Results demonstrate a functional three-column visualization system and a robust ETL workflow capable of managing complex asset lifecycles in Industry 4.0 environments.

1 Introduction

The development of a Digital Twin (DT) requires a standardized method to represent physical assets in a digital format [1]. This project implements the Asset Administration Shell (AAS) model following the IEC 63278 standard to ensure interoperability [2, 3]. The primary goal is to establish a secure, reliable information exchange layer using the BaSyx AAS Environment [4, 5].

A critical challenge in standard AAS architectures is the volatile nature of in-memory storage, which risks losing historical states during system restarts [6]. To address this, we integrate MongoDB as a persistent NoSQL layer [7]. Furthermore, the system incorporates middleware for Extract-Transform-Load (ETL) operations to bridge the gap between Operational Technology (OT) and Information Technology (IT) [8].

2 Materials and Methods

2.1 Web UI and Frontend Setup

The frontend utilizes the official Eclipse BaSyx web-ui repository [9]. Deployment involves:

- Cloning the repository and installing dependencies via `npm install` [10].
- Executing `npm run dev` to initialize the interface at `http://localhost:3000`.
- Configuring `basyx-infra.yml` to point to the backend service at port 8081.

2.2 Backend and Database Integration

The backend leverages a Docker Compose environment [11]. To enable persistence, the Java Server is configured with the `mongoDbStorage` profile [12].

```
services:
  aas-env:
    image: eclipsebasyx/aas-
        environment:2.0.0
    ports: ["8081:8081"]
    environment:
      - SPRING_PROFILES_ACTIVE=
          mongoDbStorage
      -
          SPRING_DATA_MONGODB_DATABASE
          =DT_DB
      - Basyx_Cors_Allowed-Origins
          =*
```

Listing 1: Docker Compose environment configuration

2.3 System Architecture and API Design

The system's core is the DT Data Model Building System, designed based on the BaSyx AAS Environment and compliant with the IEC 63278 standard. It acts as a middleware layer for secure and interoperable information exchange in the Industry 4.0 ecosystem. Aligned with the ISO 23247-2 reference model, it provides the following key services through BaSyx OpenAPI specifications.

2.3.1 Hierarchical Repository Services

The architecture is organized into distinct repository layers to manage the Digital Twin lifecycle from a general to a detailed level:

- **AAS Repository API:** Manages asset identification shells, supporting CRUD operations, asset information, and dynamic linking of submodels.
- **Submodel Repository API:** Manages the logical data structure, allowing hierarchical access to nested elements within complex models via `idShortPath`.
- **Concept Description Repository API:** Functions as a semantic dictionary, managing standardized definitions (e.g., ECLASS, IEC 61360) to ensure consistent technical meaning across systems.

2.3.2 Advanced Interoperability Features

To enhance flexibility and efficiency, the platform incorporates several advanced interaction mechanisms:

- **Value-Only Representation:** Endpoints ending in `/$value` are provided to transfer only the raw values of properties without metadata, minimizing data overhead for real-time updates from IoT devices.
- **Polymorphic Submodel Elements:** A wide range of element types are supported, including `Property`, `MultiLanguageProperty`, `Range`, `Blob`, and `File`, to accurately model diverse physical assets.
- **Operation Invocation:** The `/invoke` interface allows for the activation of asset functions, supporting both synchronous and asynchronous execution modes to accommodate varying task complexities.

2.3.3 Quality of Service (QoS) Management

To ensure reliability and performance in industrial settings, the system implements the following standards:

- **Standardized Result Schema:** All API responses follow a structured `Result` format, including a list of `Messages` with

clear error codes, timestamps, and severity levels (INFO, WARNING, ERROR) to facilitate system diagnostics.

- **Paging and Query Optimization:** For large data requests, the system uses cursor-based paging (`cursor`, `limit`). It also allows clients to customize the response depth (`level`) and extent (`extent`) to optimize network bandwidth.

2.4 Real-Time Data Routing

The BaSyx DataBridge connects MQTT telemetry to the AAS server [8]. We configured a `routes.json` file to map CPU and RAM usage metrics from a Python simulation script to specific Submodel elements. The workflow uses JSONata queries to extract values and HTTP PATCH requests for updates [13].

3 Results

3.1 UI Features and Flow

The interface employs a three-column layout:

- **Left:** AAS list with search and upload (`.aasx`).
- **Middle:** Hierarchical Submodel tree visualization.
- **Right:** Detailed visualizations and JSON raw data views.

3.2 Persistent Storage Results

Data is automatically organized into MongoDB collections: `assetAdministrationShells`, `submodels`, and `conceptDescriptions` [14]. This ensures data persistence across container restarts [7].

3.3 API Architecture Reference

The system maps functions according to the ISO 23247-2 model. Key schema attributes are summarized in Table 1.

Table 1: Key Schema Attributes and Technical Roles

Schema	Attribute	Purpose
AAS	`submodels`	Links Shell to functional sets.
Submodel	`elements`	Stores states and specs.
Result	`messages`	Provides status feedback.
Reference	`keys`	Defines retrieval paths.

4 Discussion

Implementation encountered 404 and CORS errors due to infrastructure misalignments [16]. These were resolved by synchronizing the UI configuration and enabling global CORS origins.

The use of Value-Only representation (`/$value`) significantly reduced network overhead. While the current setup uses Docker for demonstration, production should transition to the full Java SDK for scalability [17].

Acknowledgments

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References

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Table 2: Comprehensive API Endpoint Reference for AAS Environment [15]

Functional Group	Method	Endpoint	Description
AAS Repository	GET	`/shells`	Lists all shells.
AAS Repository	POST	`/shells`	Registers a new shell.
Submodel Repo	GET	`/submodels/{id}`	Returns submodel data.
Submodel Repo	PATCH	`/submodels/{id}/$value`	Updates values only.
Operation	POST	`/.../invoke`	Triggers functions.
File/Attachment	PUT	`/.../attachment`	Uploads file content.

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